

Compound Measures

GCSE Maths · Worked Solutions

20 questions · 62 marks total

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Section A · Foundation

Questions 1–3 are the most straightforward; they get harder as you go on. Question 10 is the most challenging.

1

[1 mark]

Complete the formula.

$$\text{speed} = \square \div \square$$

1 Speed tells you how much distance is covered in each unit of time.

2 So you divide the distance travelled by the time taken.

Answer: speed = distance ÷ time

2

[2 marks]

A car travels 180 km in 3 hours.

Work out the average speed.

$$\text{Speed} = \frac{\text{distance}}{\text{time}} = \frac{180}{3}$$

$$= 60 \text{ km/h}$$

Answer: 60 km/h

3

[2 marks]

A cyclist rides at an average speed of 15 km/h for 2 hours.

How far does the cyclist travel?

$$\text{Distance} = \text{speed} \times \text{time}$$

$$= 15 \times 2 = 30 \text{ km}$$

Answer: 30 km

4

[2 marks]

A train travels 240 km at an average speed of 80 km/h.

How long does the journey take?

$$\text{Time} = \frac{\text{distance}}{\text{speed}} = \frac{240}{80}$$

$$= 3 \text{ hours}$$

Answer: 3 hours

5

[3 marks]

A runner covers 400 m in 50 seconds.

(a) Work out the speed in m/s.

(b) Convert your answer to km/h.

$$1 \quad (a) \text{ Speed} = \frac{400}{50} = 8 \text{ m/s}$$

$$2 \quad (b) \text{ In one hour (3600 s) the runner would cover } 8 \times 3600 = 28\,800 \text{ m}$$

$$3 \quad 28\,800 \text{ m} = 28.8 \text{ km, so the speed is } 28.8 \text{ km/h}$$

$$4 \quad (\text{A useful shortcut: multiply m/s by 3.6 to get km/h.})$$

Answer: (a) 8 m/s (b) 28.8 km/h

6

[3 marks]

A block of wood has mass 480 g and volume 60 cm^3 .

Work out its density.

$$1 \quad \text{Density} = \frac{\text{mass}}{\text{volume}}$$

$$2 \quad = \frac{480}{60}$$

$$3 \quad = 8 \text{ g/cm}^3$$

Answer: 8 g/cm^3

7

[3 marks]

Gold has a density of 19.3 g/cm^3 .

Work out the mass of a piece of gold with volume 5 cm^3 .

$$1 \quad \text{Rearrange the density formula: mass} = \text{density} \times \text{volume}$$

$$2 \quad = 19.3 \times 5$$

$$3 \quad = 96.5 \text{ g}$$

Answer: 96.5 g

8

[3 marks]

A journey of 90 km is driven at an average speed of 40 km/h.

Work out the time taken, in hours and minutes.

$$1 \quad \text{Time} = \frac{90}{40} = 2.25 \text{ hours}$$

$$2 \quad \text{The } 0.25 \text{ is a fraction of an hour, not 25 minutes.}$$

$$3 \quad 0.25 \times 60 = 15 \text{ minutes}$$

Answer: 2 hours 15 minutes

9

[4 marks]

A driver travels 60 km at an average speed of 30 km/h, then 90 km at an average speed of 45 km/h.

- (a) Work out the total time for the journey.
 (b) Work out the average speed for the whole journey.

1 (a) First stage: $60 \div 30 = 2$ hours

2 Second stage: $90 \div 45 = 2$ hours, so the total time is 4 hours.

3 (b) Total distance = $60 + 90 = 150$ km

4 Average speed = $\frac{150}{4} = 37.5$ km/h

5 Note this is not the average of 30 and 45 — you must use total distance over total time.

Answer: (a) 4 hours (b) 37.5 km/h

10

[4 marks]

A box has weight 240 N. Its base measures 0.4 m by 0.3 m.

- (a) Work out the area of the base.
 (b) Work out the pressure the box exerts on the floor.
 (c) The box is turned so that a face measuring 0.4 m by 0.2 m is on the floor. Work out the new pressure.

1 (a) Area = $0.4 \times 0.3 = 0.12$ m²

2 (b) Pressure = $\frac{\text{force}}{\text{area}} = \frac{240}{0.12} = 2000$ N/m²

3 (c) New area = $0.4 \times 0.2 = 0.08$ m²

4 Pressure = $\frac{240}{0.08} = 3000$ N/m²

5 A smaller area gives a larger pressure for the same weight.

Answer: (a) 0.12 m² (b) 2000 N/m² (c) 3000 N/m²

Section B · Higher

Higher-tier style. Again, questions 1–3 are the gentlest and question 10 is the hardest.

1

[2 marks]

Convert 20 m/s into km/h.

1 In one hour there are 3600 seconds, so the distance is $20 \times 3600 = 72\,000$ m

2 $72\,000$ m = 72 km

Answer: 72 km/h

2

[2 marks]

Convert 90 km/h into m/s.

1 90 km = $90\,000$ m and 1 hour = 3600 s

2 $\frac{90\,000}{3600} = 25$

Answer: 25 m/s

3

[3 marks]

A car travels 45 miles in 50 minutes.

Work out the average speed in miles per hour.

$$1 \quad \text{The time must be in hours: } 50 \text{ minutes} = \frac{50}{60} \text{ hours} = \frac{5}{6} \text{ hours}$$

$$2 \quad \text{Speed} = 45 \div \frac{5}{6} = 45 \times \frac{6}{5}$$

$$3 \quad = 54 \text{ mph}$$

Answer: 54 mph

4

[3 marks]

A metal cube of side 4 cm has mass 1.2 kg.

Work out the density of the metal in g/cm³.

$$1 \quad \text{Volume of the cube} = 4^3 = 64 \text{ cm}^3$$

$$2 \quad \text{The units required are g/cm}^3, \text{ so convert the mass: } 1.2 \text{ kg} = 1200 \text{ g}$$

$$3 \quad \text{Density} = \frac{1200}{64} = 18.75 \text{ g/cm}^3$$

Answer: 18.75 g/cm³

5

[4 marks]

200 cm³ of liquid A with density 0.8 g/cm³ is mixed with 300 cm³ of liquid B with density 1.2 g/cm³.

Work out the density of the mixture.

$$1 \quad \text{Find each mass separately using mass} = \text{density} \times \text{volume.}$$

$$2 \quad \text{Liquid A: } 0.8 \times 200 = 160 \text{ g}$$

$$3 \quad \text{Liquid B: } 1.2 \times 300 = 360 \text{ g}$$

$$4 \quad \text{Total mass} = 520 \text{ g and total volume} = 500 \text{ cm}^3$$

$$5 \quad \text{Density} = \frac{520}{500} = 1.04 \text{ g/cm}^3$$

$$6 \quad \text{Note you cannot simply average 0.8 and 1.2, because the volumes are different.}$$

Answer: 1.04 g/cm³

6

[4 marks]

A train travels at a constant speed of 144 km/h.

How far does it travel in 25 seconds?

$$1 \quad \text{Convert the speed to m/s so it matches the time in seconds.}$$

$$2 \quad 144 \text{ km} = 144\,000 \text{ m and } 1 \text{ hour} = 3600 \text{ s}$$

$$3 \quad \frac{144\,000}{3600} = 40 \text{ m/s}$$

$$4 \quad \text{Distance} = 40 \times 25 = 1000 \text{ m}$$

$$5 \quad \text{That is } 1 \text{ km.}$$

Answer: 1000 m (1 km)

7

[4 marks]

A country has an area of $2.4 \times 10^5 \text{ km}^2$ and a population of 1.8×10^7 .

(a) Work out the population density in people per km^2 .

(b) The population increases by 20%. Work out the new population density.

1

$$\text{(a) Population density} = \frac{\text{population}}{\text{area}} = \frac{1.8 \times 10^7}{2.4 \times 10^5}$$

2

$$\text{Number parts: } 1.8 \div 2.4 = 0.75. \text{ Powers: } 10^7 \div 10^5 = 10^2$$

3

$$0.75 \times 100 = 75 \text{ people per km}^2$$

4

(b) The area does not change, so the density increases in the same proportion.

5

$$75 \times 1.2 = 90 \text{ people per km}^2$$

Answer: (a) 75 people/ km^2 (b) 90 people/ km^2

8

[4 marks]

A tap fills a 45 litre bath in 6 minutes.

(a) Work out the rate of flow in litres per minute.

(b) Work out the rate of flow in litres per second.

(c) How long would the tap take to fill a 300 litre tank?

1

$$\text{(a) } 45 \div 6 = 7.5 \text{ litres per minute}$$

2

$$\text{(b) There are 60 seconds in a minute: } 7.5 \div 60 = 0.125 \text{ litres per second}$$

3

$$\text{(c) Time} = \frac{300}{7.5} = 40 \text{ minutes}$$

Answer: (a) 7.5 litres/min (b) 0.125 litres/s (c) 40 minutes

9

[4 marks]

A cyclist rides 24 km at an average speed of 16 km/h, then returns along the same route at an average speed of 12 km/h.

Work out the average speed for the whole journey, to 2 decimal places.

1

$$\text{Outward time} = \frac{24}{16} = 1.5 \text{ hours}$$

2

$$\text{Return time} = \frac{24}{12} = 2 \text{ hours}$$

3

$$\text{Total time} = 3.5 \text{ hours and total distance} = 48 \text{ km}$$

4

$$\text{Average speed} = \frac{48}{3.5} = 13.714... \text{ km/h}$$

5

Notice this is less than the average of 16 and 12, because more time is spent at the slower speed.

Answer: 13.71 km/h (2 d.p.)

A solid metal cylinder has radius 3 cm and height 10 cm.

The metal has density 7.8 g/cm^3 and costs £0.45 per gram.

(a) Work out the volume of the cylinder, to 3 decimal places.

(b) Work out the mass of the cylinder, to the nearest gram.

(c) Work out the cost of the metal, to the nearest penny.

1 (a) Volume = $\pi r^2 h = \pi \times 3^2 \times 10 = 90\pi$

2 $90 \times 3.14159265... = 282.743 \text{ cm}^3$ (3 d.p.)

3 (b) Mass = density $\times$ volume = $7.8 \times 282.7433 = 2205.398... \text{ g}$

4 To the nearest gram that is 2205 g.

5 (c) Cost = $2205.398 \times 0.45 = 992.429...$

6 So the metal costs £992.43 (using the unrounded mass avoids rounding errors).

Answer: (a) 282.743 cm^3 (b) 2205 g (c) £992.43